Samsung Medical Center ICU Antibiotics Guide

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1. Dosing for commonly used antibiotics

Drug	Estimated glomerular filtration rate (GFR), ml/min				HD	CRRT
Acyclovir (for HSV encephalitis or VZV)	**> 50**	**10-50**	**< 10**		5 mg/kg q24hr (dose AD)	10 mg/kg q24hr
	10 mg/kg q8hr	10 mg/kg q12-24hr	5 mg/kg q24hr			
Amikacin	See Section A					
Amphotericin B (deoxycholate)	Invasive aspergillosis or mucormycosis: 1.0-1.5 mg/kg q24hr Candidemia or invasive candidiasis: 0.7-1.0 mg/kg q24hr Cryptococcal meningitis: 0.7-1.0 mg/kg IV q24hr See below for premedication and monitoring[1]					
Amphotericin B (liposomal)	3-5 mg/kg q24hr (no adjustment required) No premedication or monitoring required					
Ampicillin	**> 50**	**30-50**	**10-30**	**<10**	1-2 g q12hr (one dose AD)	1-2 g q8-12hr
	1-2 g q4-6hr	1-2 g q6-8hr	1-2 g q8-12hr	1-2 g q12hr		
Ampicillin Ampicillin/ sulbactam	3 g q6hr	3 g q8hr	3 g q12hr	3 g q24hr	3 g q24hr (dose AD)	3 g q12hr

[1] Pre-hydration: 500 ml 0.9% normal saline over 1 hr

Premedication (30-60 min before infusion): acetaminophen 650 mg PO, chlorpheniramine 1A IV, hydrocortisone 25-50 mg IV

Symptom control during infusion: antipyretics for fever, pethidine for severe rigor

Post-hydration: 500 ml 0.9% normal saline over 1 hr

Pre- and post-infusion hydration can be omitted when the risk of volume overload outweighs the benefit

Check serum electrolyte, Mg, iCa, P after infusion and replace accordingly

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Drug	Estimated glomerular filtration rate (GFR), ml/min				HD	CRRT
	> 50	**30-50**	**10-30**	**<10**		
Aztreonam	2 g q8hr	1.5 g q8hr	2 g once then 1 g q8hr	2 g q24hr	2 g q24hr AD	1.5 g q8hr
Anidulafungin	200 mg once, then 100 mg q24hr (no adjustment required)					
Caspofungin	70 mg once, then 50 mg q24hr (no adjustment required) Moderate hepatic insufficiency: 70 mg once, then 35 mg q24hr; but consider alternative echinocandins cf. Consider weight-based dosing when unresponsive to conventional regimen					
Cefazolin	**> 30**	**10-30**	**< 10**		2 g q24h	2 g q12h
	2 g q8h	2 g q12hr	2 g q24h			
Cefepime	**> 60**	**30-60**	**11-29**	**< 11**	1 g q24hr AD	2 g q12hr
	2 g q8hr	2 g q12hr	2 g q24hr	1 g q24hr		
	Administer over 4 hr					
Ceftazidime	**> 50**	**30-50**	**10-30**	**< 10**	1 g q24hr AD	2 g q8hr
	2 g q8hr	2 g q12hr	2 g q24hr	2 g q24hr		
	Administer over 4 hr					
Ceftolozane/ tazobactam (general)	**> 50**	**30-50**	**15-30**	**< 15**	0.75 g once, then 0.15 g q8hr (one dose AD)	A single LD 3.0 g (or 1.5 g q8h for 24h) then 0.75 g q8hr
	1.5 g q8hr	0.75 g q8hr	0.375 g q8hr	0.75 g once, then 0.15 g q8hr		
Ceftolozane/ tazobactam (pneumonia)	3 g q8hr	1.5 g q8hr	0.75 g q8hr	2.25 g once, then 0.45 g q8hr	2.25 g once, then 0.45 g q8hr	3 g loading, 1.5 g q8hr
Ceftriaxone	2 g q24hr (no adjustment required) 2 g q12hr for CNS infection					

Drug	Estimated glomerular filtration rate (GFR), ml/min				HD	CRRT
Cidofovir	> 55	< 55				
	5 mg/kg once weekly x2, then every other week	Contraindicated, also if Cr >1.5 mg/dL or urine protein ≥100 mg/dL				
	IV hydration: 1 L normal saline before and after cidofovir infusion Probenecid: 2 g PO 3 hr before cidofovir infusion, then 1 g at 2h and 8hr after infusion 3 mg/kg when Cr increment = 0.3-0.4 mg/dL Discontinue when Cr increment ≥0.5 mg/dL or 3+ proteinuria develops Drug acquisition: for Probenecid, Korea Orphan & Essential Drug Center (www.kodc.or.kr)					
Ciprofloxacin (general)	> 50	30-50	10-30	< 10	400 mg q24hr (dose AD)	400 mg q12hr
	400 mg q12hr	400 mg q12r	400 mg q24hr	400 mg q24hr		
Ciprofloxacin (Pseudomonas)	400 mg q8hr	400 mg q8hr	400 mg q12hr	400 mg q24hr		
Clindamycin	600-900 mg q8hr (no adjustment required) Consider 900 mg q6hr for toxic shock syndrome with necrotizing fasciitis					
Colistin[2]	CrCl (mL/min)	DOSE				
	Loading dose	300 mg 1 회 투여 12 시간후				
	≥90	150 mg q12h				
	80 to <90					
	70 to <80					
	60 to <70	130 mg q12h				
	50 to <60					
	40 to <50	100 mg q12h				
	30 to <40					
	20 to <30	80 mg q12h				
	10 to <20					
	<10	65 mg q12h				
	HD, 4 h	65 mg q12h + 50 mg after HD				
	CRRT	150 mg q8h				
Colistin (nebulized)	150 mg diluted in 10 ml NS or sterile water over 1 hr q8hr Always contact CCM, IM7, or IM3 fellow for the first order					

[2] INTERNATIONAL CONSENSUS GUIDELINES FOR THE OPTIMAL USE OF THE POLYMYXINS. Pharmacotherapy 2019;39(1):10–39.

<table>
<tr><th>Drug</th><th colspan="2">Estimated glomerular filtration rate (GFR), ml/min</th><th>HD</th><th>CRRT</th></tr>
<tr><td rowspan="2">Daptomycin</td><td>> 30</td><td>< 30</td><td rowspan="2">6-10 mg/kg q48 h (AD)
If next planned dialysis is 72 hr away, increase dose by 50%</td><td rowspan="2">6-10 mg/kg q24hr</td></tr>
<tr><td>6-10 mg/kg q24hr</td><td>6-10 mg/kg q48hr</td></tr>
<tr><td rowspan="2">Ertapenem</td><td>> 30</td><td>< 30</td><td rowspan="2">0.5 g q24hr (dose AD)</td><td rowspan="2">1 g q24hr</td></tr>
<tr><td>1 g q24hr</td><td>0.5 g q24hr</td></tr>
<tr><td rowspan="4">Fluconazole</td><td>> 50</td><td>< 50</td><td rowspan="2">Non-HD day: 200 mg q24hr
HD day: 400 mg q24hr (AD)</td><td rowspan="2">400 mg q24hr</td></tr>
<tr><td>400 mg q24hr</td><td>200 mg q24hr</td></tr>
<tr><td colspan="4">Load 800 mg for candidemia</td></tr>
<tr><td colspan="4">Double dose for certain indications (e.g., CNS infections)
For CNS infections, C. glabrata infections and some CRRT patients:
higher doses (600mg/d for normal kidney function and 800 mg/d for CRRT)</td></tr>
<tr><td>Foscarnet</td><td colspan="4"><table>
<tr><th>GFR/kg</th><th>Induction</th><th>Maintenance</th></tr>
<tr><td>> 1.4</td><td>60 mg/kg q8hr</td><td>90-120 mg/kg q24hr</td></tr>
<tr><td>1.0-1.4</td><td>45 mg/kg q8hr</td><td>70-90 mg/kg q24hr</td></tr>
<tr><td>0.8-1.0</td><td>50 mg/kg q12hr</td><td>50-65 mg/kg q24hr</td></tr>
<tr><td>0.6-0.8</td><td>40 mg/kg q12hr</td><td>80-105 mg/kg q48hr</td></tr>
<tr><td>0.5-0.6</td><td>60 mg/kg q24hr</td><td>60-80 mg/kg q48hr</td></tr>
<tr><td>0.4-0.5</td><td>50 mg/kg q24hr</td><td>50-65 mg/kg q48hr</td></tr>
<tr><td>< 0.4</td><td>Do not use</td><td>Do not use</td></tr>
</table></td></tr>
<tr><td>Ganciclovir</td><td colspan="4"><table>
<tr><th>GFR</th><th>Induction</th><th>Maintenance</th></tr>
<tr><td>> 70</td><td>5 mg/kg q12hr</td><td>5 mg/kg q24hr</td></tr>
<tr><td>50-70</td><td>2.5 mg/kg q12hr</td><td>2.5 mg/kg q24hr</td></tr>
<tr><td>25-50</td><td>2.5 mg/kg q24hr</td><td>1.25 mg/kg q24hr</td></tr>
<tr><td>10-25</td><td>1.25 mg/kg q24hr</td><td>0.625 mg/kg q24hr</td></tr>
<tr><td>< 10</td><td>1.25 mg/kg 3 times/week</td><td>0.625 mg/kg 3 times/week</td></tr>
<tr><td>HD</td><td>1.25 mg/kg 3 times/week (dose AD)</td><td>0.625 mg/kg 3 times/week (dose AD)</td></tr>
<tr><td>CRRT</td><td>2.5 mg/kg q24hr</td><td>1.25 mg/kg q24h</td></tr>
</table></td></tr>
<tr><td>Gentamicin</td><td colspan="4">See Section A</td></tr>
</table>

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Drug	Estimated glomerular filtration rate (GFR), ml/min				HD	CRRT
Imipenem/ cilastatin	**> 60**	**30-60**	**15-30**		250 mg q12h	1 g q12h **or** 1 g then 0.5 g q6h
	500 mg q6h	250 mg q6h	250 mg q8h			
	Consider higher dose below when isolate MIC 4 mcg/mL					
	> 60	**30-60**	**15-30**		500 mg q12h (AD)	1 g then 0.5 g q6h
	1 g q6h	500 mg q6h	250 mg q6h			
	ECMO dose: 750–1000 mg q6h Do not use if C_{Cr} <15 ml/min unless HD is planned within 48 hours.					
Levofloxacin	**> 50**	**20-50**	**< 20**		750 mg once, then 500 mg q48hr (dose AD)	750 mg once, then 500 mg q24hr
	750 mg q24hr	750 mg q48hr	750 mg once, then 500 mg q48hr			
Linezolid	600 mg q12hr (no adjustment required)					
Meropenem	**> 50**	**25-50**	**10-25**	**< 10**	0.5 g q24hr (dose AD)	1 g q12hr
	1 g q8hr	1 g q12hr	0.5 g q12hr	0.5 g q24hr		
	Administer over 4 hr					
	Double dose for CNS infection or higher MIC					
Metronidazole	500 mg q8hr (no adjustment required)					
Micafungin	100-150 mg q24 h (no adjustment required)					
Minocycline	200 mg once, then 100 mg q12hr (no adjustment required)					
Moxifloxacin	400 mg q24hr (no adjustment required)					
Nafcillin	1-2 g q4hr (no adjustment required)					
Oseltamivir (treatment)	**> 60**	**30-60**	**10-30**	**< 10**	30 mg once, then 30 mg AD only	No data; Consider 75 mg qd
	75 mg bid	30 mg bid	30 mg qd	No data; Consider 30 mg EOD		
Penicillin G	**> 50**	**10-50**	**< 10**		4 MU q12hr (one dose AD)	4 MU q6hr
	4 MU q4hr	4 MU q8hr	4 MU q12hr			

Drug	Estimated glomerular filtration rate (GFR), ml/min				HD	CRRT
Peramivir (for severe influenza)	**> 50**	**30-50**	**10-30**	**< 10**	100 mg once, then 100 mg 2 hr AD	No data
	600 mg q24hr	200 mg q24hr	100 mg q24hr	100 mg once, then 15 mg q24hr		
Piperacillin-tazobactam (anti-pseudomonal)	**>40**	**20-40**	**< 20**		2.25 g q8hr (0.75 g AD)	4.5 g q8hr
	4.5 g q6hr	4.5 g q8hr	2.25 g q6hr			
	Administer over 4 hr					
Teicoplanin	**> 50**	**10-50**	**< 10**		**HD**	**CRRT**
	Loading dose: 12 mg/kg q12hr x3					12 mg/kg q48hr
	12 mg/kg q24hr	12 mg/kg q48hr	12 mg/kg q72hr		12 mg/kg q72hr (dose AD)	
	Draw a trough before the 3rd maintenance dose Target trough: 20-30 mcg/ml Target trough for endocarditis: 30-40 mcg/mL					
Tigecycline	100 mg once, then 50 mg q12hr Severe hepatic insufficiency (Child-Pugh C): 100 mg, then 25 mg q12hr					
Trimethoprim/ sulfamethoxazole (for PCP or *Stenotrophomonas*)	**> 30**	**10-30**	**< 10**		Not recommended; 5-10 mg/kg/d TMP q24hr (dose AD)	5 mg/kg TMP q8hr
	15-20 mg/kg/d TMP divided q6-8hr	7.5-10 mg/kg/d TMP divided q12hr	Not recommended; 5-10 mg/kg/d TMP q24hr			
	SMC formulation: TMP/SMX IV 1 ample (5 ml) = TMP 80 mg Ex) Assume a patient with BW 60 kg and GFR 60 ml/min, then 20 mg/kg/d TMP X 60 kg X (5 ml / 80 mg) = 75 ml/d → 25 ml q8hr					

Drug	Estimated glomerular filtration rate (GFR), ml/min		HD	CRRT
Vancomycin [3] [4]	**Loading dose**: 25 mg/kg based on actual body weight **Continuous infusion** may be considered due to lower risk of nephrotoxicity (LD: 15-20 mg/kg, MD 30-40 mg/kg over 24h)			
	≥ 50	**< 50**	10 mg/kg AD	500-750 mg q12hr (effluent rate 20-25 mL/kg/h)
	15-20 mg/kg q12hr	Consult pharmacist after measuring trough level 12 hr after loading dose		
	Therapeutic drug monitoring Draw peak (1 h after end of infusion) and trough (just before administration) at 3rd dose after initiation or dosing change → Consult pharmacist for dosing			
	Target trough (for serious MRSA infections): 15-20 mcg/ml; Target AUC/MIC (for serious MRSA infections): 400-600 For continuous infusion, target Css: 20–25 mcg/mL For HD, pre-HD concentration: 15–20 mcg/ml			
Vancomycin PO for *C. difficile*	Mild/moderate: 125 mg PO q6hr Severe complicated (shock, ileus, toxic megacolon): 500 mg PO q6hr (+ metronidazole IV)			
Vancomycin enema (for *C. difficile* with ileus)	500 mg vancomycin in 100 mL normal saline per rectum q6hr as a retention enema			
Voriconazole (IV)	**> 50**	**< 50**		
	6 mg/kg q12hr x2, then 4 mg/kg q12hr	Avoid use and consider oral administration (accumulation of cyclodextran in IV formulation occurs)		
	Draw a trough 4-7 d after new dose Target trough: 1-5.5 mcg/ml			
Voriconazole (PO)	400 mg PO q12hr x2, then 200 mg PO q12hr			
	IV → PO conversion 1:1 (round to nearest tablet size)			
	Draw a trough 4-7 d after new dose Target trough: 1-5.5 mcg/ml			

[3] Katip W et al. The pharmacokinetics of vancomycin during the initial loading dose in patients with septic shock. Infect Drug Resist. 2016 Nov 22;9:253-260

[4] Reardon J et al. Vancomycin loading doses: a systematic review. Ann Pharmacother. 2015 May;49(5):557-65

A. Aminoglycosides

Once daily dosing

1) Regimens

Drug	Dose for GFR						
	> 80 (mg/kg q24hr)	60-80 (mg/kg q24hr)	40-60 (mg/kg q24hr)	30-40 (mg/kg q24hr)	20-30 (mg/kg q48hr)	10-20 (mg/kg q48hr)	0-10 (mg/kg q72hr and AD)
Amikacin	15	12	7.5	4	7.5	4	3
Gentamicin Tobramycin	5.1	4	3.5	2.5	4	3	2

2) Sampling for therapeutic drug monitoring

Peak	30 min after the completion of 30 min infusion
Trough	12 hr after start

→ Consult ICU pharmacist or order TDM

Drug	Peak	Trough
Amikacin	55-65 mcg/ml	<1 mcg/ml
Gentamicin Tobramycin	16-24 mcg/ml	<1 mcg/ml

Repeat measurement at 3rd and 7th dose, then every week or as renal function changes

Multiple daily dosing[5][6]

1) Regimens

Drug	>50	10-50	<10	HD	CRRT
Amikacin	7.5 mg/kg q12hr	7.5 mg/kg q24hr	7.5 mg/kg q48hr	7.5 mg/kg q48hr (3.25 mg/kg AD)	7.5 mg/kg q24hr
Gentamicin Tobramycin	1.7-2.0 mg/kg q8hr	1.7-2.0 mg/kg q12-24hr	1.7-2.0 mg/kg q48hr	1.7-2.0 mg/kg q48hr (0.85-1.0 mg/kg AD)	1.7-2.0 mg/kg q24hr

2) Sampling for therapeutic drug monitoring

Peak	After the completion of 3rd dose
Trough	Before 4th dose

Drug	Peak	Trough
Amikacin	15-30 mcg/ml	5-10 mcg/ml
Gentamicin Tobramycin	4-10 mcg/ml	1-2 mcg/ml

[5] An Official ATS/IDSA Statement: Diagnosis, Treatment, and Prevention of Nontuberculous Mycobacterial Diseases. Amer J Resp Crit Care Med. 2007;175(4):367–416

[6] Burton ME et al. (eds.) Applied PK & PDs: Principle of Therapeutic Drug Monitoring, 4th edition. Lippincott Williams & Wilkins, 2005.

B. Continuous infusion of beta-lactams

This section applies to continuous (over 24hr) infusion of selected beta-lactam. Thus, the duration of infusion of an infusate is equal to the interval between infusate change (i.e., 4.5 g q6hr means 4.5 g *over 6hr* q6hr).

Drug	**Loading dose**	**Maintenance dose, By GFR (ml/min)**				
Piperacillin /tazobactam	4.5 g over 30 min	**> 40**	**20-40**	**< 20**	**HD**	**CRRT**
		4.5 g q6hr	4.5 g q8hr	2.25 g q6hr	2.25 g q8hr (0.75 g AD over 30min)	4.5 g q8hr
Cefepime	1 g over 30 min	**> 60**	**30-60**	**10-29**	**<10 or HD**	**CRRT**
		2 g q8hr	2 g q12hr	2 g q24hr	1 g q24hr	2 g q12hr
Ceftazidime	1 g over 30 min	> 50	31-50	10-30	<10 or HD	CRRT
		1.5 g q6hr	1 g q6hr	0.5 g q6hr	0.25 g q6hr	1 g q6hr

2. Antibiotic lock therapy for catheter salvage in patients with catheter-related infection

A. 일반 원칙

- Antibiotic lock therapy (ALT)는 중심정맥관을 제거하기 어려운 CRBSI (catheter-related bloodstream infection) 환자에게 적용한다.
- ALT 단독 요법으로 사용해서는 안되며 반드시 전신적인 항생제 치료와 병용하여야 한다.
- 희석: 항생제는 0.9% normal saline 또는 5% dextrose water에 희석한다.
- 투여
  - 한 lumen 에 3 ml 씩, 하루에 2회 투여한다.
  - 중심정맥관으로부터 이전 체류된 용액 3 ml를 제거한 뒤 새 항생제 희석액을 주입하여야 한다.
- 금기: 항생제, 헤파린에 allergy / hypersensitivity가 있는 환자

B. 약제에 따른 투여 방법

	Vancomycin	Ciprofloxacin	Gentamicin	Amikacin
최종 농도 (mg/ml)	Vancomycin 5mg/ml, heparin 50 IU/ml	2 mg/ml	Gentamicin 2 mg/ml, heparin 50 IU/ml	2.5 mg/ml
원내 코드	XVACP	XCIPR2	XGM	XAMK
원내 제형	Vancomycin HCl 500 mg/vial (powder)	Ciprofloxacin 200 mg/100 ml (premix 제형)	80 mg/2 ml/ample	250 mg/2 ml/vial
조제 방법	① Vancomycin 0.5 g/vial (XVACP) + 0.9% normal saline 49 ml 을 혼합하여 총 용량이 50 ml 이 되도록 한다. (vancomycin 10 mg/ml) ② ① 용액에서 1.5 ml 를 취한다. ③ Heparin 1,000 IU/10 ml (XHEPA10)에서 1.5 ml 를 취한다. (heparin 150 IU) ④ ②번과 ③번을 mix 하여 total 3 ml 를 만든다	Ciprofloxacin 200 mg/100 ml (XCIPR2)에서 3 ml 를 취한다.	① Gentamicin 80 mg/2 ml/ample (XGM)중 1 ml + 0.9% normal saline 9 ml à 총 용량이 10 ml 이 되도록 한다. (gentamicin 4 mg/ml) ② ① 용액에서 1.5 ml 을 취한다. ③ Heparin 1,000 IU/10 ml (XHEPA10)에서 1.5 ml 를 취한다. (heparin 150 IU) ④ ②번과 ③번을 mix 하여 total 3 ml 를 만든다.	① Amikacin 250 mg/2 ml/vial (XAMK)중 1 ml + 0.9% normal saline 49 ml à 총 용량이 50 ml 이 되도록 한다. (amikacin 2.5 mg/ml) ② ① 용액에서 3 ml 를 취한다

C. References

1) Mermel LA, Allon M, Bouza E, et al. Clinical practice guidelines for the diagnosis and management of intravascular catheter-related infection: 2009 Update by the Infectious Diseases Society of America. Clin Infect Dis. 2009;49(1):1-45.

2) Lee MY, Ko KS, Song JH, Peck KR. In vitro effectiveness of the antibiotic lock technique (ALT) for the treatment of catheter-related infections by Pseudomonas aeruginosa and Klebsiella pneumoniae. J Antimicrob Chemother. 2007 Oct;60(4):782-7.

3) Justo JA, Bookstaver PB. Antibiotic lock therapy: review of technique and logistical challenges. Infection and Drug Resistance 2014:7 343–363

3. Intrathecal administration

	Vancomycin[1]		Gentamicin[1]		Linezolid[2]	Amphotericin B[1]
IT 투여 용량	**정상 ventricle size**	**Enlarged ventricle**	**정상 ventricle size**	**Enlarged ventricle**		
	10 mg	15 mg	3 mg	4.5 mg	10 mg	(0.01~0.5 mg) 0.1 mg
원내 코드	XVACP		XGM		XLINE	XFUNG
원내 제형	Vancomycin HCl 500 mg/vial		Gentamicin Sulfate 80 mg/2 mL/amp		Linezolid 600 mg/300 mL/bag	Amphotericin B 50 mg/vial
재구성 (액제인 경우 액제로 명시)	WI 를 가해 총 부피 10 mL 로 재구성 (재구성 후 농도: 50 mg/mL)		액제		액제	WI 를 가해 총 부피 10 mL 로 재구성 (재구성 후 농도: 5 mg/mL)
추가희석	약액 0.2 mL (10 mg) 를 뽑아 NS 2 mL 에 희석하여 총 부피 2.2 mL 로 조제	약액 0.3 mL (15 mg) 를 뽑아 NS 2 mL 에 희석하여 총 부피 2.3 mL 로 조제	① 약액 1 amp (2 mL) 을 NS 9 mL 에 희석하여 총 부피 11 mL 로 조제 ② ① 에서 0.4 mL 취하여 ③ NS 2 mL 에 추가 희석하여 총 부피 2.4 mL 로 조제	① 약액 1 amp (2 mL) 을 NS 9 mL 에 희석하여 총 부피 11 mL 로 조제 ② ① 에서 0.6 mL 취하여 ③ NS 2 mL 에 추가 희석하여 총 부피 2.6 mL 로 조제	희석 안함	2 가지 방법 **A.** 약액 1 vial (10 mL) 를 5DW 1L 에 희석한 후 2 mL 만 취하여 투여 B. 약액 0.02 mL 를 뽑아 5DW 1.98 mL 에 희석하여 총 부피 2mL 로 조제
IT 투여 형태 준비	2.2 mL	2.3 mL	2.4 mL	2.6 mL	5 mL	2 mL

	Vancomycin[1]		Gentamicin[1]	Linezolid[2]	Amphotericin B[1]
비고			1. vancomycin 과 gentamicin 은 같은 route 로 연이어 투여 가능함. 2. 주입 후 1 시간(소아 30 분) 이상 clamp 하고 필요 시 두개강 내압에 따라 clamp 시간을 조절. 3. Vancomycin 과 gentamicin 의 경우 전일 EVD 배액 된 양에 따라 다음날 IT 투여 횟수를 아래와 같이 조절. A. <50 mL/24 hours: every third day B. 50–100 mL/24 hours: every second day C. 100–150 mL/24 hours: once daily D. 150–200 mL/24 hours: increase the dosage by 5 mg of vancomycin and 1 mg of gentamicin and give once daily E. 200–250 mL/24 hours: increase the dosage by 10 mg of vancomycin and 2 mg of gentamicin and give once daily.	보통 linezolid 투여 시 IV (linezolid 600mg q12h)와 IT 를 같이 시행하는 경우가 대부분이므로 먼저 5 mL 를 IT 로 주입하고, 남은 295 mL 는 IV 로 투여함	IV 투여 병행하는 경우 위의 **A.** 방법으로 하여 IT 주고 남은 약을 IV 에 사용할 수 있음
공통	모든 IT 용 약제는 새 것을 open 하여 바로 사용 항생제 IT 투여 후 normal saline 2 ml 을 EVD 를 통해 주입				

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4. Adjuvant aerosolized amphotericin B for tracheobronchial aspergillosis

목적

- 침습적인 tracheobronchial aspergillosis에 대해 systemic antifungal과 함께 보조적으로 투여
- 미국감염학회 가이드라인에는 airway ischemia가 동반된 폐이식 환자에서 권고

  (권고 강도 강함, 근거 수준 중간)

전처치

Amphotericin B inhalation 전 short-acting $ß_2$ adrenergic receptor agonist 적용하여 bronchospasm 예방

용량 & 용법

- Amphotericin B deoxycholate 10 mg in sterile water 4 ml q12hr
  (if intolerable, amphotericin B deoxycholate 10 mg q24hr)
- Liposomal amphotericin B 50 mg in sterile water 4 ml q12hr

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5. 개정 이력

개정 시기	집필진
2018 년 12 월	감염내과 허경민 약제부 박효정, 손유민 중환자의학과 윤진구
2022 년 1 월	감염내과 허경민, 남엘리엘, 양진영, 하소영, 홍진영 약제부 박효정, 손유민, 안현영